HW09; Problem 3, Absorption Spectra

3. A portion of the solar spectrum near an absorption line has been measured.

$$\begin{aligned}
\lambda(\text{nm}) = &\{740, 741, 742, 743, 744, 745, 746, 747, 748, 749, 750, 751, 752, 753, 754, 755,\\
&756, 757, 758, 759, 760, 761, 762, 763, 764, 765, 766, 767, 768, 769, 770, 771,\\
&772, 773, 774, 775, 776, 777, 778, 779, 780\} \pm 1\\
I(\text{arbitrary units}) = &\{0.277, 0.275, 0.273, 0.271, 0.269, 0.266, 0.263, 0.259, 0.256, 0.252, 0.248, 0.244, 0.240,\\
&0.237, 0.232, 0.222, 0.202, 0.179, 0.159, 0.138, 0.120, 0.105, 0.099, 0.107, 0.119, 0.131,\\
&0.143, 0.154, 0.161, 0.166, 0.166, 0.165, 0.163, 0.160, 0.158, 0.155, 0.152, 0.149, 0.146,\\
&0.144, 0.141\} \pm 0.003
\end{aligned}$$

Treat this as a Gaussian *absorption* with a background, and analyze this data.

In[55]:=

```
lambda = Range[740, 780, 1];
i = {0.277, 0.275, 0.273, 0.271, 0.269, 0.266, 0.263, 0.259, 0.256,
   0.252, 0.248, 0.244, 0.240, 0.237, 0.232, 0.222, 0.202, 0.179, 0.159, 0.138,
   0.120, 0.105, 0.099, 0.107, 0.119, 0.131, 0.143, 0.154, 0.161, 0.166, 0.166,
   0.165, 0.163, 0.160, 0.158, 0.155, 0.152, 0.149, 0.146, 0.144, 0.141};
data = Transpose[{lambda, i}];
(*Define the model:Linear Background-Gaussian Dip*)
model = m * x + b - h * Exp[- (x - mu)^2 / (2 * sigma^2)];

(*Perform the fit with initial guesses*)
nlm = NonlinearModelFit[data, model,
   {{m, -0.003}, {b, 2.5}, {h, 0.17}, {mu, 762}, {sigma, 3}}, x];

(*Display the results*)
nlm["ParameterTable"]
Show[ListPlot[data, PlotStyle → Black, PlotLabel → "Absorbtion Fit"],
 Plot[nlm[x], {x, 740, 780}, PlotStyle → Red]]
```

Out[60]=

	Estimate	Standard Error	t-Statistic	P-Value
m	-0.00364471	0.0000424498	-85.8594	3.02537×10^{-43}
b	2.98042	0.032214	92.5192	2.0793×10^{-44}
h	0.102819	0.0016352	62.8789	2.08469×10^{-38}
mu	761.671	0.0602916	12633.1	3.01869×10^{-121}
sigma	3.34374	0.0661975	50.5116	5.07578×10^{-35}

Out[61]=